

K.Mrudula - 2520090073

1. Create Table

```
mysql> SHOW DATABASES;
+-----+
| Database |
+-----+
| bank     |
| bankdb   |
| db_2520090166 |
| information_schema |
| mysql    |
| performance_schema |
| sys      |
+-----+
7 rows in set (0.58 sec)

mysql> USE db_2520090166;
Database changed
mysql> CREATE TABLE student_marks1 (
    ->     roll_no INT PRIMARY KEY,
    ->     name VARCHAR(50),
    ->     subject VARCHAR(50),
    ->     marks DECIMAL(5,2)
    -> );
Query OK, 0 rows affected (0.26 sec)
```

2. Insert Data

```
mysql> INSERT INTO student_marks (roll_no, name, subject, marks) VALUES
    -> (1, 'Ravi', 'Math', 85.50),
    -> (2, 'Sita', 'Math', 92.75),
    -> (3, 'Anil', 'Math', 78.40),
    -> (4, 'Priya', 'Math', 88.90),
    -> (5, 'Vijay', 'Math', 80.25),
    -> (6, 'subbusir', 'aws', 98.50),
    -> (7, 'dwaraka', 'dbms', 95.75),
    -> (8, 'ranjani', 'english', 97.40),
    -> (9, 'kavithamam', 'aws1', 99.90),
    -> (10, 'seetha', 'azure', 82.25);
Query OK, 10 rows affected (0.02 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

3. Simple Select

```
mysql> SELECT * FROM student_marks;
```

roll_no	name	subject	marks
1	Ravi	Math	85.50
2	Sita	Math	92.75
3	Anil	Math	78.40
4	Priya	Math	88.90
5	Vijay	Math	80.25
6	subbusir	aws	98.50
7	dwaraka	dbms	95.75
8	ranjani	english	97.40
9	kavithamam	aws1	99.90
10	seetha	azure	82.25

```
10 rows in set (0.01 sec)
```

4. Aggregate Functions

```
mysql> SELECT COUNT(*) AS total_students FROM student_marks;
+-----+
| total_students |
+-----+
|          10 |
+-----+
1 row in set (0.02 sec)

mysql> SELECT SUM(marks) FROM student_marks;
+-----+
| SUM(marks) |
+-----+
|      899.60 |
+-----+
1 row in set (0.00 sec)

mysql> SELECT AVG(marks) FROM student_marks;
+-----+
| AVG(marks) |
+-----+
|  89.960000 |
+-----+
1 row in set (0.00 sec)

mysql> SELECT MAX(marks) FROM student_marks;
+-----+
| MAX(marks) |
+-----+
|      99.90 |
+-----+
1 row in set (0.00 sec)

mysql> SELECT MIN(marks) FROM student_marks;50
+-----+
| MIN(marks) |
+-----+
|      78.40 |
+-----+
1 row in set (0.00 sec)
```

5. Filtering with WHERE

```

MySQL 8.0 Command Line Client
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '5
SELECT * FROM student_marks WHERE marks > 85' at line 1
mysql> SELECT * FROM student_marks WHERE marks >= 90;
+-----+-----+-----+-----+
| roll_no | name      | subject | marks |
+-----+-----+-----+-----+
| 2       | Sita      | Math    | 92.75 |
| 6       | subbusir  | aws     | 98.50 |
| 7       | dwaraka   | dbms    | 95.75 |
| 8       | ranjani   | english | 97.40 |
| 9       | kavithamam | aws1    | 99.90 |
+-----+-----+-----+-----+
5 rows in set (0.01 sec)

mysql> SELECT * FROM student_marks WHERE marks < 80;
+-----+-----+-----+-----+
| roll_no | name | subject | marks |
+-----+-----+-----+-----+
| 3       | Anil | Math    | 78.40 |
+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> SELECT * FROM student_marks WHERE marks BETWEEN 80 AND 90;
+-----+-----+-----+-----+
| roll_no | name | subject | marks |
+-----+-----+-----+-----+
| 1       | Ravi  | Math    | 85.50 |
| 4       | Priya | Math    | 88.90 |
| 5       | Vijay | Math    | 80.25 |
| 10      | seetha | azure   | 82.25 |
+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> SELECT * FROM student_marks WHERE name LIKE 'P%';
+-----+-----+-----+-----+
| roll_no | name | subject | marks |
+-----+-----+-----+-----+
| 4       | Priya | Math    | 88.90 |
+-----+-----+-----+-----+
1 row in set (0.01 sec)

mysql> SELECT * FROM student_marks WHERE name IN ('Ravi', 'Sita', 'Vijay');
+-----+-----+-----+-----+
| roll_no | name | subject | marks |
+-----+-----+-----+-----+
| 1       | Ravi  | Math    | 85.50 |
| 2       | Sita  | Math    | 92.75 |
| 5       | Vijay | Math    | 80.25 |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> SELECT * FROM student_marks WHERE marks > 85 AND (subject = 'Math' OR name LIKE 'P%');

```

```

mysql> SELECT * FROM student_marks WHERE marks > 85 AND (subject = 'Math' OR name LIKE 'P%');
+-----+-----+-----+-----+
| roll_no | name | subject | marks |
+-----+-----+-----+-----+
| 1       | Ravi  | Math    | 85.50 |
| 2       | Sita  | Math    | 92.75 |
| 4       | Priya | Math    | 88.90 |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)

```

6. Update Queries

```
mysql> UPDATE student_marks SET marks = 90.00 WHERE roll_no = 3;  
Query OK, 1 row affected (0.02 sec)  
Rows matched: 1  Changed: 1  Warnings: 0  
  
mysql> UPDATE student_marks SET subject = 'Remedial Math' WHERE marks < 80;  
Query OK, 0 rows affected (0.00 sec)  
Rows matched: 0  Changed: 0  Warnings: 0  
  
mysql> _
```

7. Delete Queries

```
mysql> DELETE FROM student_marks WHERE roll_no = 5;  
Query OK, 1 row affected (0.01 sec)  
  
mysql> DELETE FROM student_marks WHERE marks < 75;  
Query OK, 0 rows affected (0.00 sec)  
  
mysql>
```

8. Sorting

```
mysql> SELECT * FROM student_marks ORDER BY marks ASC;
```

roll_no	name	subject	marks
10	seetha	azure	82.25
1	Ravi	Math	85.50
4	Priya	Math	88.90
3	Anil	Math	90.00
2	Sita	Math	92.75
7	dwaraka	dbms	95.75
8	ranjani	english	97.40
6	subbusir	aws	98.50
9	kavithamam	aws1	99.90

```
9 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM student_marks ORDER BY marks DESC;
```

roll_no	name	subject	marks
9	kavithamam	aws1	99.90
6	subbusir	aws	98.50
8	ranjani	english	97.40
7	dwaraka	dbms	95.75
2	Sita	Math	92.75
3	Anil	Math	90.00
4	Priya	Math	88.90
1	Ravi	Math	85.50
10	seetha	azure	82.25

```
9 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM student_marks ORDER BY name ASC;
```

roll_no	name	subject	marks
3	Anil	Math	90.00
7	dwaraka	dbms	95.75
9	kavithamam	aws1	99.90
4	Priya	Math	88.90
8	ranjani	english	97.40
1	Ravi	Math	85.50
10	seetha	azure	82.25
2	Sita	Math	92.75
6	subbusir	aws	98.50

```
9 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM student_marks ORDER BY name DESC;
```

roll_no	name	subject	marks
6	subbusir	aws	98.50
2	Sita	Math	92.75
10	seetha	azure	82.25
1	Ravi	Math	85.50
8	ranjani	english	97.40
4	Priya	Math	88.90
9	kavithamam	aws1	99.90
7	dwaraka	dbms	95.75
3	Anil	Math	90.00

```
9 rows in set (0.01 sec)
```

9. Group By

```
mysql> SELECT subject, SUM(marks) AS total_marks FROM student_marks GROUP BY subject;
```

subject	total_marks
Math	357.15
aws	98.50
dbms	95.75
english	97.40
aws1	99.90
azure	82.25

```
6 rows in set (0.01 sec)
```

```
mysql> SELECT subject, AVG(marks) AS avg_marks FROM student_marks GROUP BY subject;
```

subject	avg_marks
Math	89.287500
aws	98.500000
dbms	95.750000
english	97.400000
aws1	99.900000
azure	82.250000

```
6 rows in set (0.00 sec)
```

```
mysql> SELECT subject, COUNT(*) AS num_students FROM student_marks GROUP BY subject;
```

subject	num_students
Math	4
aws	1
dbms	1
english	1
aws1	1
azure	1

```
6 rows in set (0.00 sec)
```